
A Comprehensive 20-Page Technical Guide

20 Pages	10 Chapters	200+ Concepts
Fundamentals to Advanced	Algorithms & Math	Real-World Applications

Published May 2026 | For Educational Purposes

Table of Contents

Ch	Topic	Page
1	Introduction to Artificial Intelligence	3
2	Machine Learning Fundamentals	4
3	Supervised Learning	5
4	Unsupervised Learning	6
5	Neural Networks & Deep Learning	7
6	Natural Language Processing (NLP)	9
7	Computer Vision	11
8	Reinforcement Learning	13
9	ML Engineering & MLOps	15
10	Ethics, Trends & Future of AI	17
—	Appendix: Algorithms Cheat Sheet	19
—	Glossary of Key Terms	20

Chapter 1: Introduction to Artificial Intelligence

What is Artificial Intelligence?

Artificial Intelligence (AI) is the simulation of human intelligence processes by computer systems. These processes include learning (acquiring information and rules), reasoning (using rules to reach conclusions), and self-correction. AI was formally founded as a discipline at the Dartmouth Conference in 1956 by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon.

Core Branches of AI

Branch	Description	Example
Machine Learning	Systems that learn from data	Spam filters
Deep Learning	Multi-layer neural networks	Image recognition
NLP	Processing human language	ChatGPT, Siri
Computer Vision	Interpreting visual input	Facial recognition
Robotics	Physical AI agents	Boston Dynamics
Expert Systems	Rule-based reasoning	Medical diagnosis

The AI Hierarchy

AI is the broadest field, encompassing any technique that enables machines to mimic human intelligence. Machine Learning is a subset of AI that focuses on data-driven learning. Deep Learning is a subset of ML that uses artificial neural networks with many layers. This hierarchy is foundational to understanding how modern AI systems are built.

Key Insight: Narrow AI (ANI) solves specific tasks; General AI (AGI) would match human cognition across domains; Super AI (ASI) would surpass human intelligence — only ANI exists today.

Chapter 2: Machine Learning Fundamentals

Defining Machine Learning

Machine Learning is a method of data analysis that automates analytical model building. Based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention. Arthur Samuel (1959) defined ML as a field that gives computers the ability to learn without being explicitly programmed.

The ML Workflow

Step	Activity	Tools
1. Data Collection	Gather raw data from sources	APIs, Web scraping, Sensors
2. Data Preprocessing	Clean, normalize, encode	Pandas, NumPy, Scikit-learn
3. Exploratory Analysis	Visualize patterns & distributions	Matplotlib, Seaborn, Plotly
4. Feature Engineering	Create/select relevant features	Feature tools, PCA
5. Model Selection	Choose algorithm	Scikit-learn, PyTorch
6. Training	Fit model to training data	GPU clusters, cloud
7. Evaluation	Measure performance on test data	Confusion matrix, ROC
8. Deployment	Serve model in production	Docker, FastAPI, AWS

Key Mathematical Concepts

- **Linear Algebra:** Vectors, matrices, dot products, eigenvalues — the language of ML computations.
- **Calculus:** Derivatives and gradients used in optimization (gradient descent).
- **Probability & Statistics:** Distributions, Bayes' theorem, hypothesis testing.
- **Optimization:** Minimizing loss functions using algorithms like SGD, Adam, RMSProp.

Chapter 3: Supervised Learning

Overview

Supervised learning uses labeled training data — input/output pairs — to learn a mapping function. The goal is to predict outputs for new, unseen inputs. It is the most widely used ML paradigm, covering two primary tasks: Classification (discrete output) and Regression (continuous output).

Key Algorithms

Algorithm	Type	Strengths	Weaknesses
Linear Regression	Regression	Simple, interpretable	Assumes linearity
Logistic Regression	Classification	Probabilistic output	Linear boundaries only
Decision Trees	Both	Interpretable, fast	Prone to overfitting
Random Forest	Both	High accuracy, robust	Less interpretable
SVM	Both	Effective in high dims	Slow on large data
Gradient Boosting	Both	State-of-the-art accuracy	Slow to train
KNN	Both	Simple, no training phase	Slow at inference
Naive Bayes	Classification	Fast, good for text	Feature independence

Evaluation Metrics

- **Accuracy:** Fraction of correct predictions. Simple but misleading on imbalanced data.
- **Precision & Recall:** Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$. Trade-off matters.
- **F1-Score:** Harmonic mean of precision and recall — balanced metric.
- **ROC-AUC:** Area under the ROC curve — measures classifier ranking ability.
- **MSE / RMSE:** Mean Squared Error for regression — penalizes large errors.
- **R-squared:** Proportion of variance explained by the model.

Chapter 4: Unsupervised Learning

What is Unsupervised Learning?

Unsupervised learning finds hidden patterns in data without labeled responses. The system tries to discover the underlying structure of data on its own. It is particularly useful for exploratory analysis, data compression, and anomaly detection.

Clustering Algorithms

Algorithm	Approach	Best Used For
K-Means	Centroid-based	Spherical, similarly-sized clusters
DBSCAN	Density-based	Arbitrary shapes, noise detection
Hierarchical	Agglomerative/Divisive	Unknown number of clusters
Gaussian Mixture	Probabilistic	Soft cluster assignments
Mean Shift	Mode-seeking	Finding density peaks

Dimensionality Reduction

High-dimensional data is hard to visualize and computationally expensive. Dimensionality reduction compresses data while retaining the most important information.

- **PCA (Principal Component Analysis):** Linear projection maximizing variance. Fast and widely used.
- **t-SNE:** Non-linear technique for 2D/3D visualization of high-dimensional data.
- **UMAP:** Faster than t-SNE, better preserves global structure. Popular in genomics.
- **Autoencoders:** Neural network-based compression; learns compact latent representations.

Association Rule Mining

Discovers interesting relations between variables in large datasets. The classic example is market basket analysis: 'customers who buy bread also buy butter.' The Apriori algorithm and FP-Growth are the dominant approaches, evaluated using Support, Confidence, and Lift.

Chapter 5: Neural Networks & Deep Learning

Biological Inspiration

Artificial Neural Networks (ANNs) are loosely inspired by the human brain's structure. The brain contains ~86 billion neurons connected by synapses. An artificial neuron (perceptron) receives inputs, applies weights, sums them, then passes through an activation function to produce an output.

Activation Functions

Function	Formula	Range	Use Case
Sigmoid	$1/(1+e^{-x})$	(0, 1)	Binary classification output
Tanh	$\tanh(x)$	(-1, 1)	Hidden layers (zero-centered)
ReLU	$\max(0, x)$	[0, inf)	Most hidden layers — fast
Leaky ReLU	$\max(0.01x, x)$	(-inf, inf)	Avoids dying ReLU
Softmax	$e^x / \sum(e^x)$	(0,1) sum=1	Multi-class output
GELU	$x * \Phi(x)$	(-inf, inf)	Transformers (BERT, GPT)

Backpropagation & Gradient Descent

Training a neural network involves forward propagation (computing outputs), calculating a loss (error), then backpropagation — computing gradients of the loss with respect to each weight using the chain rule, and updating weights to minimize loss. Variants include SGD, Momentum, AdaGrad, RMSProp, and Adam.

Deep Learning Architectures

Architecture	Abbreviation	Primary Application
Feedforward Network	FNN / MLP	Tabular data, classification
Convolutional Neural Network	CNN	Images, video, spatial data
Recurrent Neural Network	RNN	Sequences, time-series
Long Short-Term Memory	LSTM	Long sequences, NLP
Gated Recurrent Unit	GRU	Faster alternative to LSTM
Transformer	—	NLP, vision, multimodal AI
Generative Adversarial Network	GAN	Image synthesis, data augmentation

Variational Autoencoder	VAE	Generative modeling, compression
Diffusion Model	—	Image/audio/video generation

Chapter 5 (continued): Transformers & Modern Deep Learning

The Transformer Architecture

Introduced in the landmark 2017 paper 'Attention Is All You Need' by Vaswani et al., the Transformer uses self-attention mechanisms to process all tokens in parallel, overcoming the sequential bottleneck of RNNs. It consists of encoder and decoder stacks with multi-head attention, positional encoding, feed-forward layers, and layer normalization.

Regularization Techniques

- **Dropout:** Randomly deactivates neurons during training to prevent co-adaptation.
- **Batch Normalization:** Normalizes layer inputs — accelerates training, reduces sensitivity.
- **L1 / L2 Regularization:** Adds weight penalty to loss — discourages large weights.
- **Early Stopping:** Halts training when validation loss stops improving.
- **Data Augmentation:** Expands training data via transforms (flips, crops, noise).

Hyperparameter Tuning

Critical hyperparameters include learning rate, batch size, number of layers and neurons, dropout rate, and optimizer type. Tuning strategies include grid search, random search, Bayesian optimization (Optuna), and Neural Architecture Search (NAS). Learning rate scheduling (warmup, cosine decay) is essential for stable training.

Rule of Thumb: Learning rate is the most important hyperparameter. Start with 1e-3 for Adam, use learning rate finders, and always monitor training vs. validation loss curves.

Chapter 6: Natural Language Processing (NLP)

Introduction to NLP

Natural Language Processing is the branch of AI that enables computers to understand, interpret, and generate human language. NLP bridges linguistics, computer science, and ML. Applications range from simple text classification to sophisticated conversational AI systems capable of nuanced reasoning and generation.

NLP Pipeline

Stage	Description	Example Output
Tokenization	Split text into tokens	['Hello', ',', 'world']
Normalization	Lowercase, remove noise	'hello world'
Stop word removal	Remove common words	Remove 'the', 'is'
Stemming/Lemmatization	Reduce to root form	'running' -> 'run'
POS Tagging	Label grammatical role	NN, VB, JJ, RB
NER	Identify named entities	Apple=ORG, London=LOC
Parsing	Analyze sentence structure	Dependency / constituency
Embedding	Map words to vectors	Word2Vec, GloVe, BERT

Word Representations

- **Bag of Words (BoW):** Simple word count vector — ignores order and semantics.
- **TF-IDF:** Weights words by frequency in document vs. corpus — better than BoW.
- **Word2Vec:** Trains shallow network to predict context — dense semantic vectors.
- **GloVe:** Global co-occurrence statistics — fast, pre-trained embeddings available.
- **FastText:** Sub-word embeddings — handles morphology and rare words.
- **BERT / RoBERTa:** Contextual embeddings — same word gets different vector per context.

Chapter 6 (continued): Large Language Models

The Rise of Large Language Models

Large Language Models (LLMs) are transformer-based models trained on massive text corpora. They exhibit emergent capabilities — abilities not explicitly programmed — at scale. GPT-4, Claude, Gemini, LLaMA, and Mistral represent the current state of the art, capable of reasoning, code generation, summarization, translation, and creative writing.

Model	Organization	Parameters	Key Feature
GPT-4o	OpenAI	~200B (est.)	Multimodal, instruction-tuned
Claude 3.5/4	Anthropic	~100B+ (est.)	Safety, long context
Gemini Ultra	Google	~1T (est.)	Natively multimodal
LLaMA 3	Meta	8B–405B	Open weights
Mistral Large	Mistral AI	~70B	Efficient, MoE variants
Phi-3	Microsoft	3.8B–14B	Small but powerful

Fine-tuning Strategies

- **Full Fine-tuning:** Update all model weights — expensive but maximally flexible.
- **LoRA (Low-Rank Adaptation):** Inject trainable low-rank matrices — 10-100x fewer parameters.
- **QLoRA:** Quantized LoRA — fine-tune 65B models on a single GPU.
- **Prompt Tuning:** Learn soft prompt embeddings — model weights frozen.
- **RLHF:** Reinforcement Learning from Human Feedback — aligns LLMs with human preferences.
- **DPO:** Direct Preference Optimization — simpler RLHF alternative.

Chapter 7: Computer Vision

What is Computer Vision?

Computer Vision (CV) is the field of AI that trains computers to interpret and understand visual information from images and videos — replicating human visual perception. It combines digital image processing, machine learning, and deep learning to enable machines to see, identify, track, and respond to visual inputs.

Core CV Tasks

Task	Description	Example Models
Image Classification	Assign a label to an image	ResNet, EfficientNet, ViT
Object Detection	Locate & classify objects	YOLO, Faster R-CNN, DETR
Semantic Segmentation	Label every pixel by class	FCN, DeepLab, SegFormer
Instance Segmentation	Separate individual objects	Mask R-CNN, SAM
Pose Estimation	Detect keypoints on body	OpenPose, MediaPipe
Optical Flow	Estimate motion between frames	FlowNet, RAFT
Depth Estimation	Predict depth from 2D image	MiDaS, DepthPro
Image Generation	Create new images	Stable Diffusion, DALL-E 3

Convolutional Neural Networks in Depth

CNNs extract hierarchical features: early layers detect edges and textures, middle layers detect parts (wheels, eyes), and deeper layers detect complex objects. Key components include convolutional layers (learnable filters), pooling layers (spatial downsampling), and fully connected layers (classification head).

Landmark CNN Architectures

Architecture	Year	Innovation
LeNet-5	1998	First successful CNN for digit recognition
AlexNet	2012	Deep CNN; won ImageNet by large margin
VGGNet	2014	Very deep networks with 3x3 convolutions
GoogLeNet/Inception	2014	Inception modules; 1x1 convolutions
ResNet	2015	Residual/skip connections; 152 layers

EfficientNet	2019	Compound scaling — SOTA efficiency
Vision Transformer (ViT)	2020	Patch-based transformers for images
ConvNeXt	2022	CNN that matches ViT performance

Chapter 7 (continued): Advanced Vision Techniques

Data Augmentation for Vision

- **Geometric transforms:** Random flips, rotations, crops, shearing.
- **Color jitter:** Brightness, contrast, saturation, hue variations.
- **CutOut / Random Erasing:** Mask random image regions during training.
- **Mixup & CutMix:** Blend two images and their labels.
- **AutoAugment / RandAugment:** Learn/randomize augmentation policies.

Transfer Learning in Computer Vision

Pre-trained models on ImageNet (or CLIP, DINOv2) provide powerful feature extractors. Strategies: (1) Feature extraction — freeze backbone, train only classifier head. (2) Fine-tuning — unfreeze and retrain all or part of backbone with small LR. (3) Zero-shot — use CLIP embeddings with text prompts, no task-specific training.

Generative Models in Vision

Diffusion models have surpassed GANs for high-fidelity image generation. They work by learning to reverse a gradual noising process. Stable Diffusion, DALL-E 3, Midjourney, and Imagen are prominent examples. ControlNet adds spatial conditioning (edges, poses, depth maps) for precise control over generation.

Foundation Models: SAM (Segment Anything Model) by Meta segments any object with minimal prompting. DINOv2 produces universal visual features without supervision. These represent the shift toward universal vision models.

Chapter 8: Reinforcement Learning

RL Fundamentals

Reinforcement Learning (RL) trains agents to make decisions by rewarding desired behaviors and/or penalizing undesired ones. Unlike supervised learning, there is no labeled dataset — the agent learns through trial-and-error interaction with an environment. RL powers game-playing AI (AlphaGo, AlphaZero), robotics, autonomous vehicles, recommendation systems, and LLM alignment (RLHF).

Concept	Definition
Agent	The learner / decision maker
Environment	Everything the agent interacts with
State (s)	Current situation representation
Action (a)	Choice made by the agent
Reward (r)	Feedback signal from environment
Policy (π)	Mapping from states to actions
Value Function	Expected cumulative future reward
Q-Function	Value of taking action a in state s
Episode	Complete sequence from start to terminal state
Discount (γ)	How much future rewards are discounted

Major RL Algorithms

Algorithm	Type	Key Idea
Q-Learning	Value-based	Learn Q-values via Bellman equation
DQN	Value-based	Q-Learning with deep NN function approximator
REINFORCE	Policy gradient	Optimize policy directly via MC returns
A2C / A3C	Actor-Critic	Parallel workers, value baseline
PPO	Policy gradient	Clipped objective for stable updates
SAC	Off-policy AC	Maximum entropy — exploration + exploitation
DDPG / TD3	Deterministic AC	Continuous action spaces
AlphaZero	Model-based	MCTS + self-play — superhuman games

Chapter 8 (continued): Deep RL & Applications

Exploration vs. Exploitation

A central dilemma in RL: the agent must balance exploiting known high-reward actions vs. exploring unknown actions that might yield higher rewards. Strategies include epsilon-greedy (random action with probability epsilon), Boltzmann exploration (softmax over Q-values), UCB (upper confidence bound), and Thompson sampling.

Real-World RL Applications

- **Game Playing:** AlphaGo defeated world champion Lee Sedol (2016). AlphaZero mastered Chess, Shogi, Go from self-play in 24 hours.
- **Robotics:** OpenAI's Dactyl learned dexterous manipulation; Boston Dynamics uses RL for locomotion.
- **Autonomous Driving:** RL for decision-making in complex traffic scenarios.
- **Healthcare:** RL for personalized treatment planning and drug dosage optimization.
- **Finance:** Algorithmic trading, portfolio optimization, and market-making.
- **LLM Alignment:** RLHF and RLAIFF align LLM behavior with human preferences.

Multi-Agent RL (MARL)

MARL studies multiple agents interacting in shared environments. Challenges include non-stationarity (other agents change the environment), credit assignment, and emergent communication. AlphaStar (StarCraft II) and OpenAI Five (Dota 2) are landmark MARL achievements using population-based training.

Key Challenge: Sample efficiency — RL often requires millions of environment interactions. Model-based RL (learning an environment model to plan) and imitation learning (learning from expert demonstrations) help address this.

Chapter 9: ML Engineering & MLOps

The ML Engineering Gap

Only ~5-10% of real ML systems code is the actual model. The majority consists of data pipelines, feature stores, serving infrastructure, monitoring, CI/CD, and configuration management. MLOps (Machine Learning Operations) bridges the gap between model development and reliable production deployment.

The ML System Stack

Layer	Components	Example Tools
Data Ingestion	Collection, streaming, batch	Kafka, Airflow, Spark
Feature Engineering	Transform, validate, store	Feast, Tecton, dbt
Model Training	Distributed training, experiment tracking	PyTorch, MLflow, W&B
Model Registry	Version, stage, lineage	MLflow, DVC, Neptune
Model Serving	REST API, batch, streaming	TorchServe, Triton, BentoML
Monitoring	Drift, performance, data quality	Evidently, Fiddler, Arize
CI/CD	Test, build, deploy pipelines	GitHub Actions, Jenkins, Tekton
Orchestration	Pipeline scheduling & management	Kubeflow, Prefect, ZenML
Infrastructure	Compute, storage, networking	AWS, GCP, Azure, K8s

Model Deployment Patterns

- **REST API:** Synchronous, request-response. Simple and universal. FastAPI, Flask.
- **Batch Inference:** Process large datasets offline on schedule. Cost-efficient.
- **Streaming Inference:** Real-time predictions on data streams. Kafka + model server.
- **Edge Deployment:** Run models on devices (phones, IoT). TFLite, CoreML, ONNX Runtime.
- **Shadow Deployment:** Run new model in parallel without serving its predictions — safely test.
- **A/B Testing:** Route traffic splits between model versions — measure impact.
- **Canary Deployment:** Gradually shift traffic to new model — catch regressions early.

Chapter 9 (continued): Monitoring & Model Governance

Data & Model Drift

Deployed models degrade as the world changes. Data drift occurs when the input distribution shifts from the training distribution. Concept drift occurs when the relationship between inputs and outputs changes. Both require statistical monitoring (PSI, KL divergence, Kolmogorov-Smirnov test) and automated retraining triggers.

Key MLOps Metrics

Metric	Description	Target
Model Latency (p50/p99)	Inference response time	< 100ms (p50)
Throughput (QPS)	Queries per second	Per SLA
Prediction Accuracy	Live model performance	Within 5% of offline
Data Freshness	Age of features at inference	< 1 hour (streaming)
Training Frequency	How often model retrains	Weekly / trigger-based
Model Availability	Uptime of serving endpoint	> 99.9%
Feature Coverage	% of requests with all features	> 98%

Model Quantization & Optimization

- **FP32 -> FP16 / BF16:** Halves memory — negligible accuracy loss on most models.
- **INT8 Quantization:** 4x memory reduction vs FP32 — requires careful calibration.
- **Pruning:** Remove low-importance weights — reduces size and compute.
- **Knowledge Distillation:** Train small 'student' model to mimic large 'teacher'.
- **ONNX:** Cross-framework model exchange format — enables hardware-specific optimization.

Chapter 10: Ethics, Safety & the Future of AI

AI Ethics Principles

Principle	Description
Fairness	Models must not discriminate based on protected attributes
Transparency	Decisions should be explainable to affected parties
Accountability	Clear responsibility for AI system outcomes
Privacy	Data minimization; right to erasure; GDPR compliance
Beneficence	AI should benefit individuals and society
Non-maleficence	Prevent harm — do not deploy systems that could cause harm
Autonomy	Respect human agency and avoid manipulative design
Robustness	Systems should be reliable and secure under adversarial conditions

Bias in AI Systems

Bias enters ML systems through biased training data, biased problem framing, and biased evaluation. Facial recognition systems have shown higher error rates for darker skin tones. Hiring algorithms have discriminated against women. Bias mitigation strategies include diverse data collection, fairness constraints during training, and adversarial debiasing.

AI Safety Research

- **Alignment:** Ensuring AI systems pursue goals intended by designers. Hardest unsolved problem.
- **Interpretability / XAI:** LIME, SHAP, attention visualization — understanding model decisions.
- **Robustness:** Adversarial training, certified defenses against adversarial examples.
- **Constitutional AI:** Anthropic's approach — train AI using a set of principles.
- **Red-teaming:** Stress-test AI systems for harmful behaviors before deployment.

Chapter 10 (continued): AI Trends & Future Outlook

Current Trends (2024-2025)

Trend	Description
Multimodal AI	Models processing text, image, audio, video simultaneously
Agentic AI	AI that takes multi-step actions using tools to complete tasks
On-device AI	Running LLMs on phones, laptops — privacy, latency benefits
Test-time Compute	Spending more compute at inference for better reasoning (o1, R1)
Open Source Models	LLaMA, Mistral, Phi — closing gap with proprietary models
AI Coding Assistants	GitHub Copilot, Cursor — transforming software engineering
Synthetic Data	Generating training data using AI — scales beyond human annotation
Mixture of Experts (MoE)	Sparse models activating only relevant sub-networks

Regulatory Landscape

- **EU AI Act (2024):** Risk-based regulation — prohibits certain uses, mandates transparency for high-risk AI.
- **US Executive Order on AI (2023):** Safety standards, watermarking requirements, export controls.
- **China AI Regulations:** Content generation registration, algorithmic recommendation rules.
- **GDPR:** Gives EU citizens rights regarding automated decisions affecting them.

The Road Ahead

AI is progressing rapidly along multiple fronts simultaneously. Scaling laws suggest continued improvement with more compute and data. Architectural innovations (sparse models, state space models, neuromorphic computing) may unlock new efficiency frontiers. The most important questions are not technical but societal: who benefits, who is harmed, who decides, and how do we ensure AI remains aligned with human values and flourishing.

Key Prediction: By 2030, AI is expected to contribute \$15.7 trillion to the global economy (PwC). The most impactful applications will be in healthcare diagnostics, scientific discovery, climate modeling, and personalized education.

Appendix: ML Algorithms Cheat Sheet

Algorithm	Type	Data Size	Interpretable	Training Speed	Best For
Linear Regression	Supervised	Any	Yes	Very Fast	Continuous target, linear
Ridge / Lasso	Supervised	Any	Yes	Very Fast	Regularized regression
Logistic Regression	Supervised	Any	Yes	Fast	Binary classification
Decision Tree	Supervised	Medium	Yes	Fast	Quick baselines
Random Forest	Supervised	Medium-Large	Partial	Medium	Tabular, robust baseline
Gradient Boosting	Supervised	Medium-Large	Partial	Slow	Best tabular accuracy
XGBoost / LightGBM	Supervised	Large	Partial	Medium	Competitions, tabular
SVM	Supervised	Small-Medium	No	Slow	High-dim, small data
KNN	Supervised	Small	Yes	None	Simple, no training needed
K-Means	Unsupervised	Any	Yes	Fast	Spherical clustering
DBSCAN	Unsupervised	Medium	Yes	Medium	Noise-robust clustering
PCA	Unsupervised	Any	Partial	Fast	Dimensionality reduction
Neural Network	Supervised	Large	No	Slow	Images, NLP, audio
CNN	Supervised	Large	No	Slow (GPU)	Images, video
RNN/LSTM	Supervised	Large	No	Slow	Sequences, time-series
Transformer	Supervised	Very Large	No	Very Slow	NLP, vision, multimodal
Naive Bayes	Supervised	Any	Yes	Very Fast	Text classification

Glossary of Key AI / ML Terms

Term	Definition
Accuracy	Proportion of correct predictions out of all predictions.
Attention Mechanism	Weights inputs by relevance — core of Transformer architecture.
Autoencoder	Neural network that learns compressed latent representations.
Backpropagation	Algorithm to compute gradients via the chain rule for weight updates.
Batch Size	Number of samples processed before updating model weights.
Bias (ML)	Error from incorrect assumptions — underfitting. Also: neuron offset term.
Classification	Predicting a discrete class label from inputs.
Cross-Entropy	Loss function for classification — measures divergence from truth.
Data Augmentation	Artificially increasing training data via transforms.
Dropout	Regularization: randomly zero out neuron activations during training.
Embedding	Dense vector representation of discrete objects (words, items).
Epoch	One complete pass through the entire training dataset.
Gradient Descent	Iterative optimization: move weights in direction of negative gradient.
Hyperparameter	Configuration set before training (LR, depth, dropout rate).
Inference	Using a trained model to make predictions on new data.
Loss Function	Measures prediction error — the objective to minimize.
Overfitting	Model memorizes training data — poor generalization to new data.
Precision	$TP / (TP + FP)$ — how many positive predictions were correct.
Recall	$TP / (TP + FN)$ — how many actual positives were caught.
Regularization	Techniques to reduce overfitting (L1, L2, Dropout, Early Stopping).
Regression	Predicting a continuous numerical value from inputs.
Reinforcement Learning	Agent learns via reward/penalty feedback from an environment.
Transfer Learning	Reuse a pre-trained model's learned representations for a new task.
Underfitting	Model too simple to capture data patterns — high bias.
Variance (ML)	Sensitivity to training data fluctuations — overfitting tendency.